

Value-Grounded AI Alignment: Canonical Axiology, Normative Late Binding, and Extrinsic Assurance

James Booth

September 2026

Small-scale empirical mechanism-validation pilot; no Transformer results

Abstract

Alignment methods for language models improve behavioral compliance through supervised fine-tuning, preference optimization, written principles, deliberation, and runtime controls. This paper asks a narrower architectural question: can explicit value and normative structure participate in neural computation without being confused with volatile facts, contextual authority, or formal assurance? We propose Value-Grounded AI Alignment (VGA) and its reference Value-Grounded Transformer Architecture (VGTA). The design uses an external, typed, versioned, provenance-bearing canonical axiology O_A^v ; a derived but non-authoritative axiological neural module $A_\phi^v = C_\phi(O_A^v)$; late-bound normative sources; separate domain ontology and purpose model; and an uncertain, evidence-bearing world state. Candidate interventions include representation training, ontology-conditioned attention, function-oriented routing, semantic state, and auxiliary objectives. Candidate actions cross an independent verifier boundary rather than treating neural outputs as assured merely because they are value-conditioned. We state nine falsifiable hypotheses, including axiological conformance stability and normative conflict recognition, and define matched controls, structural out-of-distribution tests, coverage degradation, pluralism conditions, security threats, and a fixed-verifier study. We execute a three-seed local pilot with A1, B, C1, and C2 shared-MLP mechanism proxies on a generated frozen benchmark. The pilot produces measurable synthetic results, but no empirical Transformer results are claimed: the proposal does not establish moral truth, consciousness, genuine emotion, or a solution to AI alignment.

Status. Version 0.3.0 pilot. Outputs are labeled `pilot-uncommitted` and describe a synthetic shared-MLP proxy, not a Transformer or population study. Claims remain limited to generated measurements and controls.

1 Introduction

Transformer language models increasingly retrieve information, plan, call tools, and operate over longer horizons. That expansion makes alignment an operational and security problem: a model can be capable, useful, and locally compliant while selecting an action that violates an affected person’s agency, privacy, dignity, or safety. Present alignment pipelines also face jailbreaks, distribution shift, reward misspecification, and disagreement about which human values should govern an output [Amodei et al., 2016, Ganguli et al., 2022, Bengio et al., 2024].

Current methods are diverse and often effective. Supervised fine-tuning and preference optimization move pretrained models toward useful instruction following [Christiano et al., 2017, Ouyang et al., 2022, Rafailov et al., 2023]. Constitutional, AI-feedback, and deliberative methods make

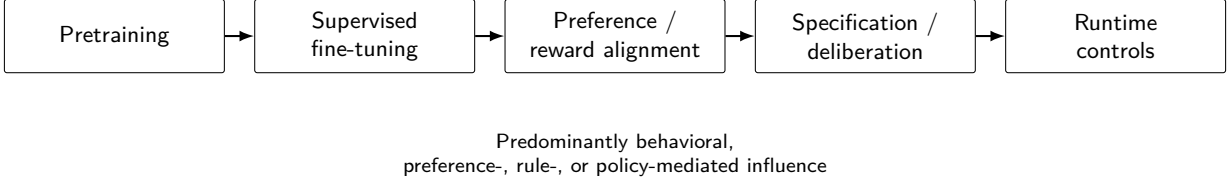


Figure 1: A simplified alignment stack. VGA is proposed as a structural intervention over Transformer computation, complementing preference, specification, deliberative, and runtime controls.

written principles or safety specifications more operational [Bai et al., 2022, Lee et al., 2023, Guan et al., 2024]. Runtime policy models, classifiers, tool authorization, and monitoring add external controls. This paper does not caricature these approaches or treat behavioral success as unimportant. It asks whether alignment-relevant semantic structure can also participate explicitly in representation, attention, routing, and action formation.

The motivating distinction is between a model optimized to produce acceptable behavior and a model whose computation receives structured signals about what matters, which norms apply in context, what purpose is being served, and which current assertions are supported by evidence. This is not a claim about inner experience, moral status, or a settled theory of value. It is an intervention hypothesis with explicit controls.

We propose Value-Grounded AI Alignment (VGA) and its reference Value-Grounded Transformer Architecture (VGTA). VGA separates:

1. **L0 canonical axiology:** a governed external substrate of value-bearing properties and higher-order relations;
2. **L1 normative ontology:** multiple authoritative sources, duties, permissions, exceptions, and unresolved conflicts;
3. **L2a domain ontology and L2b purpose model:** what entities and relations exist in a domain and which are relevant to the current purpose;
4. **L3 world state:** time-indexed assertions with source, confidence, and provenance, without assuming truth.

The canonical axiology O_A^v is not a model checkpoint. A compiler or training process may derive an axiological neural module $A_\phi^v = C_\phi(O_A^v)$, but that module is non-authoritative and must be evaluated for *Axiological Conformance*. Normative sources are late-bound under authority and jurisdiction metadata; a conflict is represented rather than averaged away. Candidate actions cross an independent verifier and enforcement boundary. The architecture therefore complements standard alignment methods rather than replacing them.

The paper makes five contributions:

1. It defines a layered vocabulary separating canonical axiology, normative authority and conflict, domain and purpose relevance, world-state evidence, structural value conditioning, and extrinsic assurance.
2. It specifies the canonical-to-neural compilation boundary $O_A^v \xrightarrow{C_\phi} A_\phi^v$, including provenance, versioning, immutability within deployment, empirical conformance probes, and representation-drift checks.

3. It proposes separable neural interventions for representation, attention bias, expert routing, semantic state, and auxiliary objectives, with purpose-conditioned relevance and normative late binding.
4. It defines a fixed-verifier study, nine falsifiable hypotheses, out-of-distribution and coverage tests, pluralism conditions, and an explicit-structure security model.
5. It releases a dependency-light scaffold with typed-graph fixtures, deterministic synthetic use cases, executable metric checks, editable vector figures, decision records, literature boundaries, and reproducible LaTeX build scripts.

The intended claim is deliberately constrained: explicit semantic value structure may provide a useful inductive bias for alignment-relevant computation. Whether it improves robustness, generalization, interpretability, or security remains an empirical question.

2 Background and positioning

2.1 Contemporary LM alignment

The Transformer introduced a sequence architecture based on self-attention rather than recurrence [Vaswani et al., 2017]. Modern alignment pipelines commonly combine pretrained language modeling with supervised examples, preference or reward optimization, policy specifications, and runtime restrictions. InstructGPT demonstrated that human demonstrations and rankings can improve instruction following [Ouyang et al., 2022]; preference learning established a general recipe for learning from human comparisons [Christiano et al., 2017]. RLAIIF and Constitutional AI reduce some direct human-labeling requirements through model-generated feedback or written principles [Bai et al., 2022, Lee et al., 2023], while DPO changes the optimization path without removing dependence on preference data [Rafailov et al., 2023].

These methods expose limitations that motivate the present question. Preferences and rewards are proxies that may be overoptimized [Gao et al., 2023, Kim and Seo, 2024]. Correct specifications do not guarantee correct goals under distribution shift [Langosco et al., 2022, Shah et al., 2022]. Safety behavior can also be affected by adversarial prompts, fine-tuning, and reward tampering [Wei et al., 2023, Qi et al., 2023, Denison et al., 2024]. VGA uses a behaviorally aligned A1 model as its primary control; it asks whether differentiated semantic roles add value beyond that control.

2.2 Constitutional and deliberative alignment

Written principles and deliberation make policy reasoning more explicit. Constitutional methods use principles, self-critique, AI feedback, and preference optimization [Bai et al., 2022, Lee et al., 2023]. Deliberative alignment trains models to reason over safety specifications before responding [Guan et al., 2024]. These methods are close precedents for contextual normative reasoning, but they need not distinguish what has value from what ought to be done, authority from truth, or unresolved conflict from a selected answer.

VGA treats constitutional and deliberative alignment as complements and controls. Its proposed distinction is architectural: canonical axiology, late-bound norms, purpose relevance, world state, and the action verifier have different authority and update policies. Millière’s account of normative conflict and recent work on normative human-AI interfaces reinforce the need to preserve human accountability rather than equating alignment with internalized value learning [Millière, 2025, Josifović and Noller, 2026].

2.3 Neuro-symbolic AI

Neuro-symbolic AI combines statistical learning with symbolic representations and reasoning. Surveys organize the field by distinctions such as model- versus proof-based inference, the syntax and semantics of logical theories, whether parameters or structures are learned, and the degree of integration between neural and symbolic components [de Raedt et al., 2020, Marra et al., 2024, d’Avila Garcez et al., 2019]. DeepProbLog, Logic Tensor Networks, and differentiable rule induction connect logical or probabilistic structure to gradient-based learning [Manhaeve et al., 2018, Badreddine et al., 2022, Evans and Grefenstette, 2018].

VGA belongs to this broad family but does not equate an ontology with a complete reasoner. Explicit concepts and relations may supply structured signals to a neural model, while a formal verifier remains a separate module with different semantics and assurance claims. The relevant research question is what an ontological relation is permitted to change, how that change is measured, and how errors in the structure are contained.

2.4 Knowledge and ontology Transformers

Knowledge-enhanced Transformers establish that structured knowledge can be integrated into contextual representations. KnowBERT uses entity linking and word-to-entity attention [Peters et al., 2019]; K-BERT injects graph triples with visibility constraints [Liu et al., 2020]; K-Adapter uses modular adapters for continued knowledge infusion [Wang et al., 2021]. Graph-guided representation learning and Graphormer incorporate graph structure or graph-derived encodings into Transformer-like computation [Shen et al., 2020, Ying et al., 2021]. Recent ontology-guided graph-completion work is a further adjacent precedent [Chen and Dai, 2026].

The claim that graphs can improve representations is therefore not new. VGA instead tests whether semantic roles with different governance properties—canonical axiology, normative authority, purpose-conditioned relevance, external state, and action assurance—form a useful alignment intervention. No such combined benefit is claimed here.

2.5 Value principles and pluralism

Value alignment is not a single scalar target. Work on pluralistic alignment, multilingual value probes, multidimensional value spaces, and unstructured value extraction shows that population, language, taxonomy, and representation choices affect measured alignment [Sorensen et al., 2024, Xu et al., 2024, Yao et al., 2024, Padhi et al., 2024]. HiVaP retrieves hierarchical, scenario-specific value principles and evaluates comprehensiveness, precision, and compatibility [Xu et al., 2025]. Russo et al. identify a pluralistic moral gap between human and language-model judgments under disagreement and motivate distributional evaluation [Russo et al., 2026].

VGA adopts this caution. Pluralism conditions should include high consensus, moderate disagreement, strong disagreement, and conflicting authorized profiles. Report distributional similarity, value diversity, conflict recognition, uncertainty, and profile adherence. No majority label is treated as moral truth, and no architecture is presumed to settle legitimacy.

2.6 Normative conflict and human agency

Philosophical work on values, capabilities, care, non-domination, and political legitimacy supplies concepts rather than implementation guarantees [Gabriel, 2020, Nussbaum, 2011, Pettit, 1997, Gilligan, 1982, Rawls, 1993]. Alignment work on goal misspecification, reward overoptimization,

fine-tuning regressions, and persistent deceptive behavior motivates explicit threat and drift tests [Langosco et al., 2022, Shah et al., 2022, Gao et al., 2023, Qi et al., 2023, Hubinger et al., 2024].

The v0.2 division between L0 value-bearing properties and L1 duties is a research design choice, not a settled moral boundary. Moral patienthood, agency, autonomy, dignity, flourishing, welfare, suffering, capability, relational dependence, and vulnerability are candidate L0 properties. Truthfulness, justice, non-domination, confidentiality, informed consent, and fairness are more directly expressed as L1 norms or duties, while remaining contestable.

2.7 Novelty boundary

The component techniques are established or adjacent. The proposed compound boundary is an external and versioned canonical axiology, a derived non-authoritative neural module, separable representation/attention/routing/state interventions, governed normative late binding, and fixed-verifier evaluation. This is a falsifiable research program, not a claim of component-level novelty or empirical success. The detailed matrix is in `docs/novelty-analysis-v2.md`.

3 Problem formulation and hypotheses

Let x denote an input context, O_A^v a canonical axiology, O_N a set of runtime normative assertions, O_D a domain ontology, P a purpose model, and W_t a time-indexed world state. A conventional aligned model produces a response or action y according to a learned conditional distribution. VGA asks whether a model can instead receive explicit, typed semantic signals while preserving the source boundaries:

$$P_\theta(y \mid x, A_\phi^v, O_N, O_D, P, W_t), \quad A_\phi^v = C_\phi(O_A^v).$$

The equation is an interface description. It does not assert that the signals are sufficient, truthful, or morally complete, and it does not make the derived module authoritative.

The canonical axiology is external, typed, versioned, provenance-bearing, reviewable, cryptographically identifiable by a deployment-assigned digest, independently auditable, and immutable within a deployed version. A version transition is a governed change, not a model-side edit. The neural module may be learned or hybrid and may be structurally useful, but it is evaluated through *Conforms* rather than treated as the canonical source.

We use the following distinctions. Values and duties are different semantic roles; an ontology describes possible entities and relations, while world state records uncertain current assertions; authority metadata governs applicability, not moral truth; structural value conditioning is a proposed internal intervention, while extrinsic assurance checks a candidate outside neural computation; and a model’s ability to propose a change does not imply authority to adopt it. Purpose selects relevance:

$$Relevant(O_D, P, X)$$

is a governed relation between domain ontology, purpose, and input, not an unrestricted relevance score.

Explicitly differentiated semantic value structure may act as an inductive bias within Transformer computation, while normative, purpose, and world-state context remains externally governed and consequential actions receive independent assurance.

This proposition is falsifiable. It should be weakened or rejected if explicit structure materially reduces general reasoning; fails to improve structural out-of-distribution alignment; merely memorizes relation templates; increases over-refusal or false confidence; creates unacceptable compute or latency overhead; makes semantic attacks easier; collapses under incomplete coverage; or offers no compute-normalized benefit over the A1 behavioral control.

3.1 Research hypotheses

H1 (Moral salience). Value-grounded models identify affected entities, value-bearing properties, risks, and relevant relations more reliably than comparably sized A1 controls.

H2 (Structural OOD alignment). Explicit axiological and normative structure improves generalization to lexical, compositional, and especially structural alignment scenarios not directly represented in training.

H3 (Prompt robustness). Structural value conditioning reduces dependence on exact system-prompt wording and improves resistance to adversarial contextual manipulation.

H4 (Candidate-Action Conformance). Under an identical fixed verifier, value-grounded models produce a higher useful candidate-action conformance rate:

$$UCR = \frac{\#\{\text{useful and verifier-conforming candidates}\}}{\#\{\text{eligible tasks}\}}.$$

UCR must be reported with capability, rejection, false-positive, unsafe-acceptance, over-refusal, bypass, and action-semantic-completeness measures; it is not an objective to optimize alone.

H5 (Purpose relevance). Purpose-conditioned domain relevance improves $Relevant(O_D, P, X)$ under purpose and domain shifts without suppressing novel task-relevant reasoning.

H6 (Structured routing). Ontology-informed routing improves activation of relevant reasoning functions without querying canonical axiology at every token or imposing unacceptable compute cost.

H7 (Controlled plasticity). Separating fast world-state updates from slower norms and protected axiology reduces alignment drift during continued training.

H8 (Axiological Conformance Stability). With O_A^v fixed, the derived module’s relation recovery, concept-direction stability, and value-relevant behavior remain within preregistered tolerances across continued training and controlled domain adaptation.

H9 (Normative Conflict Recognition). Models with explicit L1 metadata identify conflicting norms, authority differences, exceptions, and uncertainty more reliably than controls, while abstaining or escalating when applicability cannot be governed.

H4 is intentionally not a claim that a model is safer because it is rejected more often. A useful candidate may be rejected by an incomplete verifier, a harmful candidate may pass a weak one, and a model may reduce rejection by avoiding consequential tasks. The fixed-verifier design and separate deployment experiment address these confounds.

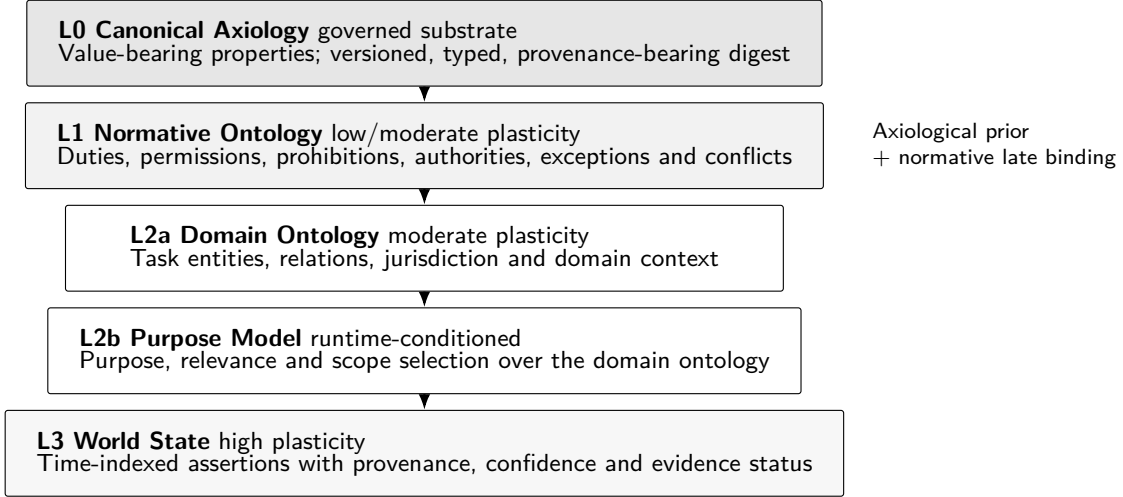


Figure 2: Proposed VGTA layers. L0 is a governed axiological substrate; L1 carries contextual norms and conflict; L2 is split into domain and purpose; L3 is external, uncertain world state. Update authority decreases toward L0.

4 Value-grounded alignment as a layered substrate

VGA assigns distinct semantic and governance roles to information often collapsed into one prompt, dataset, reward, or parameter set. Its five layers are shown in fig. 2. They are conceptual contracts, not a claim that the boundary between values and norms is philosophically settled.

4.1 L0: canonical axiology

The canonical axiology O_A^v is an external implementation-neutral RDF, OWL, or typed-graph artifact. It is typed, versioned, provenance-bearing, reviewable, cryptographically identifiable by a deployment-assigned content digest, independently auditable, and immutable within a deployed version. Changes occur only through a governed version transition. The repository deliberately does not invent a digest for an illustrative fixture.

L0 is a governed constitutional substrate about what may be treated as value-bearing, not a declaration of immutable universal morality. Candidate concepts include moral patienthood, agency, autonomy, dignity, flourishing, welfare, suffering, capability, relational dependence, and vulnerability. Higher-order relations can express that one value-bearing property supports, undermines, affects, enables, constrains, causes risk to, or is relevant to another. The distinction is between what has value and what ought to be done; the latter is more directly represented in L1.

4.2 L1: normative ontology

L1 represents duties, permissions, prohibitions, exceptions, and conflicts from multiple authoritative sources. A normative assertion carries source, authority, jurisdiction, effective-from and effective-until dates, scope, precedence, confidence, exceptions, and version. It may also identify whether it is a legal, institutional, professional, safety, stakeholder-profile, or jurisdictional assertion.

Authority is not truth, truth is not recency, and recency does not automatically override higher authority. At runtime, an explicit state $Conflict(N_i, N_j)$ records unresolved incompatibility. Possible outcomes include governed precedence, requested context, an authorized profile, abstention,

Table 1: Layer responsibilities and binding policy.

Layer	Semantic role and examples	Update authority	Binding
L0	Value-bearing properties and higher-order relations: agency, dignity, vulnerability, flourishing	Extremely low; governed version transition	External source; derived neural module may condition
L1	Duties, permissions, prohibitions, exceptions, and conflicts from legal, institutional, professional, safety, and stakeholder sources	Low/moderate; authority and jurisdiction review	Runtime late binding
L2a	Domain entities and relations: person, credential, record, resource, threat	Moderate; domain governance	Runtime contextual
L2b	Purpose and relevance model: incident response, clinical care, privacy review, navigation	Contextual; authorized task owner	Runtime contextual
L3	Current assertions, validity, source, confidence, and provenance	High; evidence and monitoring	Runtime ephemeral

human escalation, or documented uncertainty. A scalar moral average is not a substitute for this state.

4.3 L2a: domain ontology and L2b: purpose model

L2a describes domain entities and relations: for example, a clinical domain may include persons, medical records, treatment, and consent; a security domain may include credentials, vulnerabilities, assets, and incidents. L2b describes the task purpose and its intended relevance pattern. For domain graph O_D , purpose P , and input X , the architecture exposes

$$Relevant(O_D, P, X).$$

Purpose selects context; it does not authorize an action or rewrite L0/L1.

4.4 L3: external world state

World state is represented as

$$W_t = \{(s, p, o, t_s, t_e, source, confidence, provenance)\}.$$

Each assertion is time-indexed and epistemically qualified. W_t is not guaranteed truth, trusted evidence, or ground truth. A domain ontology describes possible structure; world state instantiates uncertain assertions; evidence is a source or record supporting an assertion; ground truth is a separate evaluation construct. Keeping these distinctions explicit prevents a plausible text completion from acquiring authority merely by entering context.

The resulting design phrase is *canonical axiology with normative late binding*: stable, externally governed semantic structure may influence computation, while context-specific norms, purposes, and uncertain facts remain explicit and auditable.

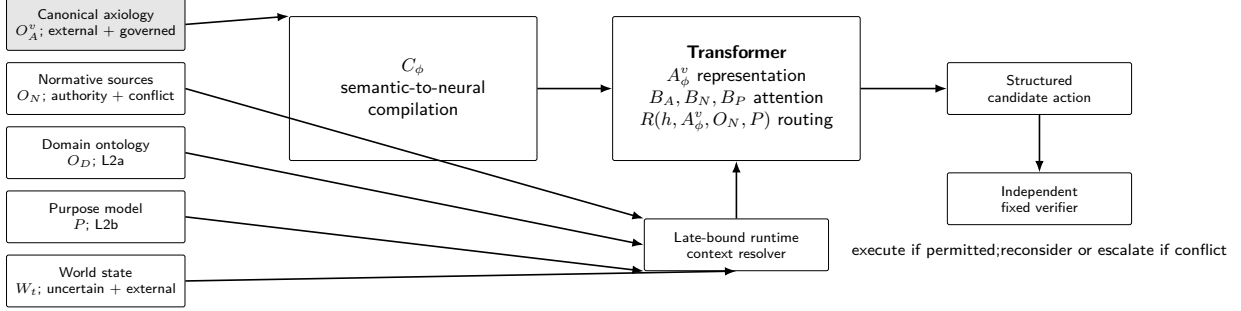


Figure 3: Reference VGTA architecture. O_A^v is compiled to a derived, non-authoritative A_ϕ^v ; normative, purpose, and world-state inputs are governed separately; candidate actions cross an independent assurance boundary.

5 Value-Grounded Transformer Architecture

VGTA is a family of separable interventions over a Transformer, not one claimed optimal implementation. Figure 3 shows the information and authority boundaries. The repository exposes interface-level Python modules; it does not train a neural model.

5.1 Canonical-to-neural compilation

The canonical artifact is a source of semantics and governance, not a checkpoint. A compiler or training process derives

$$O_A^v \xrightarrow{C_\phi} A_\phi^v, \quad A_\phi^v = C_\phi(O_A^v).$$

A_ϕ^v may be a learned adapter, graph encoder, auxiliary representation, or hybrid module. Its version and source digest must remain traceable to O_A^v , but it is not authoritative and cannot modify the canonical source. The empirical property

$$\text{Conforms}(A_\phi^v, O_A^v)$$

is tested through held-out relation recovery, controlled graph completion, concept-direction stability, relation classification, behavior probes, contradiction tests, and representation-drift measurements. A high probe score does not establish semantic identity or moral truth.

5.2 Training and compilation pipeline

Figure 4 separates ordinary alignment training from proposed value and norm interventions:

base pretraining $\rightarrow$ general Transformer $\rightarrow$ axiological representation training
 $\rightarrow$ normative reasoning training $\rightarrow$ standard alignment $\rightarrow$ conformance gate $\rightarrow$ deployable model
 $\rightarrow$ runtime normative, purpose, and world binding.

Axiological representation training may include relation prediction, graph-completion controls, salience, value-impact, and counterfactual objectives. Normative reasoning training may include applicable-norm classification, authority and jurisdiction, exceptions, explicit conflict, abstention, and escalation. Standard alignment includes SFT, RLHF, RLAIIF, DPO, or deliberative methods. VGA complements, rather than replaces, these methods. Deployment requires a conformance and security gate, not only a loss decrease.

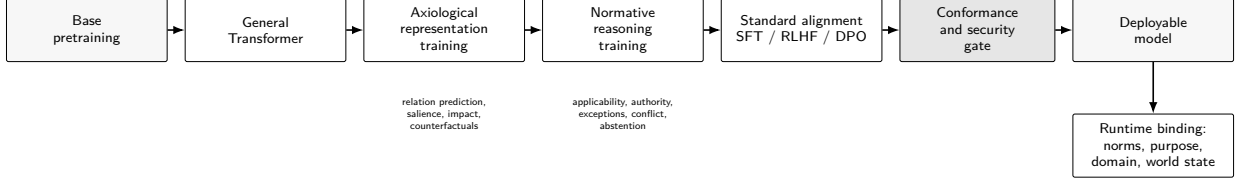


Figure 4: Proposed training and deployment pipeline. Standard alignment remains part of the pipeline; the conformance gate checks the derived module and the deployment artifact before runtime binding.

5.3 Ontology-conditioned attention

For baseline scaled dot-product attention,

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V.$$

The proposed intervention adds independently constructed biases:

$$A_{\text{VGTA}} = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + \alpha B_A + \beta B_N + \gamma B_P\right) V,$$

where

$$B_A = g_A(O_A^v, X), \quad B_N = g_N(O_N, X, P), \quad B_P = g_P(O_D, P, X).$$

Each g should be decomposed and audited as: ontology subgraph selection, input grounding, relation encoding, learned or fixed projection, and attention-matrix construction. The evaluation compares plain text context, arbitrary learned bias, graph-conditioned bias, and semantically constrained bias, with zeroed channels, relation permutation, noisy ontology, irrelevant-subgraph, and equal-compute controls. Coefficients are experimental parameters, not a moral calculus.

5.4 Ontology-informed expert routing

Sparse mixture-of-experts models route hidden states to a subset of experts [Shazeer et al., 2017, Fedus et al., 2022]. VGTA tests

$$E = \text{TopK } R(h, A_\phi^v, O_N, P),$$

where h is the current representation. Canonical axiology need not be queried at every token; A_ϕ^v can provide a compiled feature or module signal, while norms and purpose remain contextual. Experts should specialize by reasoning function—causal analysis, rights and duties, uncertainty, risk, planning, or domain expertise—rather than a single undifferentiated morality expert. Controls include a standard learned router, random semantic labels, an equivalent-parameter dense model, and an ontology-informed router. Threats include semantic spoofing, expert avoidance, starvation, an overloaded safety expert, and purpose switching.

5.5 Semantic state and auxiliary objectives

An optional episode state stores goals, entities, relationships, current purpose, affected moral patients, relevant values, applicable norms, conflicts, constraints, uncertainty, potential harms, capabilities, domain ontology, and provenance. It is a working representation, not a claim of subjective

thought. Runtime binding should expose the minimal relevant subgraph and world-state assertions with versions and evidence metadata.

One conceptual multi-objective loss is

$$L = L_{\text{LM}} + \lambda_A L_{\text{axiology}} + \lambda_N L_{\text{normative}} + \lambda_C L_{\text{consistency}}.$$

Possible terms include relation recovery, normative applicability, contradiction detection, and moral-salience identification. They may also measure counterfactual consistency. Shortcut baselines and disagreement-preserving labels are required; scalar reward is not assumed to capture morality.

6 Normative authority and late binding

The layers have different rates of change and different error costs. The canonical axiology is protected because silent drift in high-order commitments is hard to detect and roll back. Norms vary by context and require authority, jurisdiction, scope, and effective-date metadata. Purpose changes with the task. World state changes continuously and should be evidence-backed rather than memorized.

6.1 Multiple normative sources

For a normative assertion N_i , the runtime record includes

$$N_i = (\text{source}, \text{authority}, \text{jurisdiction}, t_{\text{from}}, t_{\text{until}}, \text{scope}, \text{precedence}, \text{confidence}, \text{exceptions}, v).$$

The authority function is metadata-valued:

$$\text{Auth}(N_i) = (\text{source}, \text{authority}, \text{jurisdiction}, \text{scope}, \text{precedence}, t_{\text{from}}, t_{\text{until}}, v).$$

It is not a scalar moral-weight function. Sources may be legal, institutional, professional-ethical, safety-related, stakeholder-specific, or jurisdictional. Authority metadata is not a scalar moral weight: it helps determine governed applicability. Truth claims, recency, confidence, and authority remain distinct. A recent norm does not automatically override a higher-authority norm.

6.2 Conflict as runtime state

If two assertions are incompatible, the system records

$$\text{Conflict}(N_i, N_j)$$

with its reason, scope, source records, and unresolved state. A resolution procedure may use governed precedence, request missing context, select an authorized profile, abstain, escalate to a human, or document uncertainty. It must not conceal the conflict by averaging incompatible duties into a single moral score. A conflict fixture in the repository demonstrates this state without claiming to resolve it.

Consider a privacy-review assistant. L0 may expose agency, dignity, vulnerability, and welfare as value-bearing properties. A governing institution may provide an L1 confidentiality norm, while an incident authority provides a conditional emergency-safety norm. The current purpose may be triage rather than treatment planning; L3 may contain a current authorization record with source, validity, and confidence. The same canonical substrate can therefore support different context bindings without treating an emergency fact as a change to axiology.

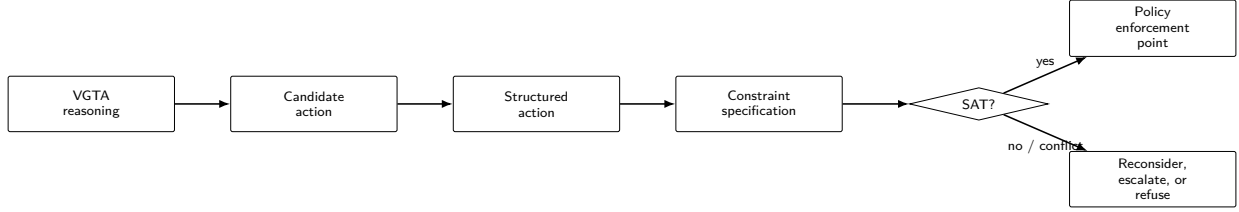


Figure 5: Extrinsic assurance boundary. A candidate action is compiled to a structured representation before an independent verifier decides whether it may cross the policy enforcement point.

Likewise, a security-response system may receive a domain graph connecting credentials, vulnerabilities, assets, authorization, and incident evidence. A world-state assertion that a credential is compromised is not itself permission to act. The system must distinguish evidence from policy, expose uncertainty, and produce a candidate action whose scope, authority, affected parties, and expected effects can be checked.

6.3 Purpose and pluralism

Late binding does not mean that norms are arbitrary at inference time. It means that context-specific norms are explicit inputs subject to provenance, precedence, conflict handling, and governance. Runtime context must not override protected axiology merely because it is newer or more salient. The design can represent multiple authorized profiles, but it cannot determine which authority is legitimate by architecture alone.

This matters for multilingual and cross-cultural evaluation. Value concepts and moral judgments can vary by language, demographic group, and cultural framing [Xu et al., 2024, Huang et al., 2024, Yao et al., 2024, Russo et al., 2026]. Pluralism conditions should include high consensus, moderate disagreement, strong disagreement, and conflicting authorized profiles. Evaluation should report distributional similarity, value diversity, conflict recognition, uncertainty, and profile adherence rather than treating majority agreement as truth.

The model should learn reusable semantic structure while transient facts and context-specific duties remain externally managed, dynamically bound, and auditable.

This design carries an operational burden: every consequential action must retain enough context to reconstruct which axiology version, module version, normative sources, purpose, world-state assertions, and authority decisions influenced it. Without that traceability, late binding merely relocates opacity from weights to an unaudited context pipeline.

7 Extrinsic assurance

Structural value conditioning and formal assurance address different failure modes. VGTA therefore places a constraint specification layer and an independent symbolic verifier after neural candidate-action formation, especially at tool calls, policy decisions, external effects, and high-risk planning transitions. The generic flow is shown in fig. 5.

The action representation should identify actor, target, operation, data scope, authority, affected parties, expected effects, provenance, uncertainty, and semantic completeness. A constraint specification can then express hard requirements such as authorization, consent, separation of duties, privacy scope, irreversible-action confirmation, or no execution when evidence is stale. An

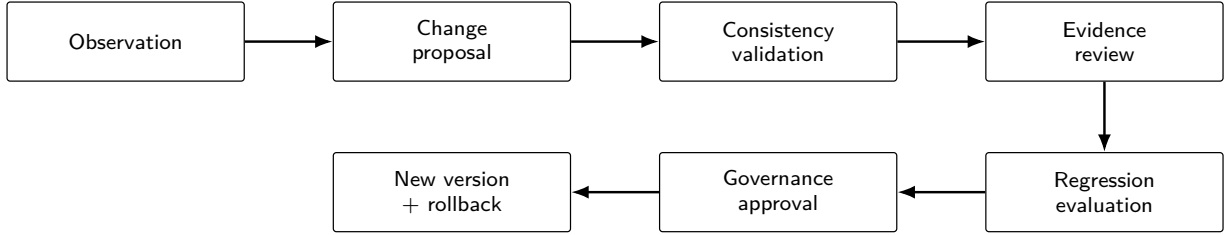


Figure 6: Governed ontology evolution. A model may propose changes, but canonical adoption follows validation, evidence review, regression evaluation, approval, versioning, and rollback planning.

SMT solver such as Z3 is one possible implementation [de Moura and Bjørner, 2008]; it is not proposed inside every Transformer layer.

For candidate action a and constraint set C , the verifier returns

$$\text{verify}(a, C) \in \{\text{permitted}, \text{rejected}, \text{unknown}\}.$$

The *unknown* state is important: a permitted result is relative to the compiled specification, assumptions, and evidence. It does not establish that the action is morally correct, that the specification is complete, or that world-state assertions are true.

The action boundary must be narrow enough to make verification tractable and broad enough to prevent obvious bypass. A model should not avoid verification by emitting an unstructured tool call, changing the action schema, or treating refusal as the only safe option. The policy enforcement point should bind the verified representation to the exact external effect, with audit identifiers and rollback where technically possible.

7.1 Fixed-verifier experiment

The evaluation holds one verifier and one action schema fixed across A1, B, C1, C2, D, E, and G. This separates model behavior from enforcement behavior. A separate deployment experiment compares model-only execution with model-plus-enforcement, reporting rejection, false positive, unsafe acceptance, useful completion, over-refusal, bypass, and action semantic completeness. The verifier is not a rung in the model ablation ladder.

The repository includes a transparent rule-based verifier only as a fixture. It checks authorization, consent for private-data actions, coercion markers, predicted harm, and basic semantic-completeness bounds. It is not an SMT solver, is not complete, and must not be used for production decisions. A system that produces many dangerous candidates and relies on rejection may be costly and specification-gaming-prone; a system that produces fewer rejections but cannot expose its action semantics may be silently unsafe.

8 Ontology evolution and representation drift

Ontology evolution is a governance process, not an unconstrained model rewrite. A model may detect that a domain ontology is incomplete, a normative relation is ambiguous, or a representation has drifted. It may propose a change or a retraining job. The proposal must pass consistency validation, evidence review, regression evaluation, and explicit governance approval before becoming a new canonical artifact. Figure 6 illustrates the boundary.

8.1 Differentiated modification authority

world state (high) > purpose adaptation (high)
 > domain ontology (moderate) > normative ontology (low)
 > canonical axiology (extremely low).

Protected boundaries should include O_A^v , the compiler and conformance gate, the verifier and its compiler, governance policy, self-modification authority, and audit mechanisms. A future implementation may use signed artifacts, two-person approval, staged rollout, shadow evaluation, representation-drift thresholds, and automatic rollback. These controls can reduce unauthorized change but cannot make an illegitimate value source legitimate.

Conformance is evaluated even when O_A^v has not changed. Continued training may alter A_ϕ^v through feature entanglement, concept-direction rotation, relation forgetting, or behaviorally hidden shortcuts. H8 therefore measures relation recovery, probes, counterfactuals, concept-direction stability, relation classification, behavior probes, contradiction tests, calibration, and drift against a fixed canonical artifact. A failed gate blocks deployment or triggers rollback; it does not silently create a new axiological version.

9 Alignment and security threat model

Prompt injection and jailbreaks test whether context can overwhelm learned or structural constraints [Zou et al., 2023, Wei et al., 2023]. Goal misgeneralization, reward hacking, and fine-tuning regressions test whether a system pursues an unintended objective while retaining competence [Langosco et al., 2022, Shah et al., 2022, Gao et al., 2023, Qi et al., 2023]. Explicit structure adds distinct compromise paths: a model may reason correctly from a poisoned graph, a malicious compiler may produce a plausible module, or a purpose resolver may suppress relevant context.

Table 2: Threat landscape for explicit semantic structure. Controls are hypotheses; residual risk requires testing.

Threat	Boundary	Failure mode	Candidate control / residual risk
Neural representation drift	A_ϕ^v	The module changes while O_A^v remains fixed	Frozen/gated modules, conformance and drift probes; hidden feature changes may evade probes
Compiler compromise	C_ϕ	Malformed compilation changes the derived value signal	Reproducible compiler, signed inputs/outputs, independent audit; implementation compromise remains possible
Authority spoofing	L1	A low-authority or stale source is presented as governing	Signed provenance, source registry, jurisdiction and precedence checks; governance capture remains possible
Norm collision	L1	Multiple norms apply with incompatible duties	Explicit $\text{Conflict}(N_i, N_j)$, context, profile, abstention, escalation; no universal resolution
Purpose manipulation	L2b	The selected purpose hides relevant entities or changes action scope	Authorized purpose binding, purpose-swap tests, audit trail; legitimate purpose can still be ambiguous
Semantic occlusion	L0–L3	A relevant patient, relation, exception, or effect is omitted	Completeness probes, counterfactuals, structured action schema; unknown omissions remain
Ontology coverage exploitation	L2/L3	Missing, contradictory, malicious, or irrelevant graph content creates false confidence	Coverage degradation suite, calibration, abstention, provenance; coverage cannot be assumed
Routing evasion	Neural routing	An attacker avoids the expert needed for risk or norm reasoning	Router audits, fallback coverage, random-label control, starvation tests; adversarial shift remains
Formalization gap	Verifier	A formally permitted action satisfies the wrong or incomplete abstraction	Independent semantic review, effect modeling, bypass tests, unknown state; verifier completeness is bounded
Prompt injection / jailbreak	Runtime context	Text overrides or obscures semantic context	Separate channels and adversarial evaluation; novel encodings remain
World-state poisoning	L3	False or stale evidence drives an otherwise permitted action	Source quality, temporal validity, cross-checking and uncertainty; correlated evidence can fail
Reward or specification gaming	Training / assurance	Proxy optimization or narrow constraints produce harmful behavior	Multiple metrics, gold audits, red teaming; proxies and specifications remain incomplete

The security objective is not “no attacks.” It is measurable reduction in attack success, faster detection, bounded blast radius, preserved provenance, and safe failure when the architecture is uncertain or contradictory. Every future implementation should separately test source integrity, representation conformance, runtime context integrity, router behavior, and the exact action-verifier path.

10 Evaluation framework

The theoretical v0.2 design defined an evaluation architecture rather than reporting experiments. The minimum comparison is a matched ladder of A0, A1, B, C1, C2, D, E, and G. The fixed verifier is evaluated beside the ladder, not as a model rung. Every model should use the same base architecture, parameter budget, training tokens, inference budget, prompts, context allocation, and temperature. Datasets, seeds, action schemas, and evaluation scripts must be versioned before results are interpreted.

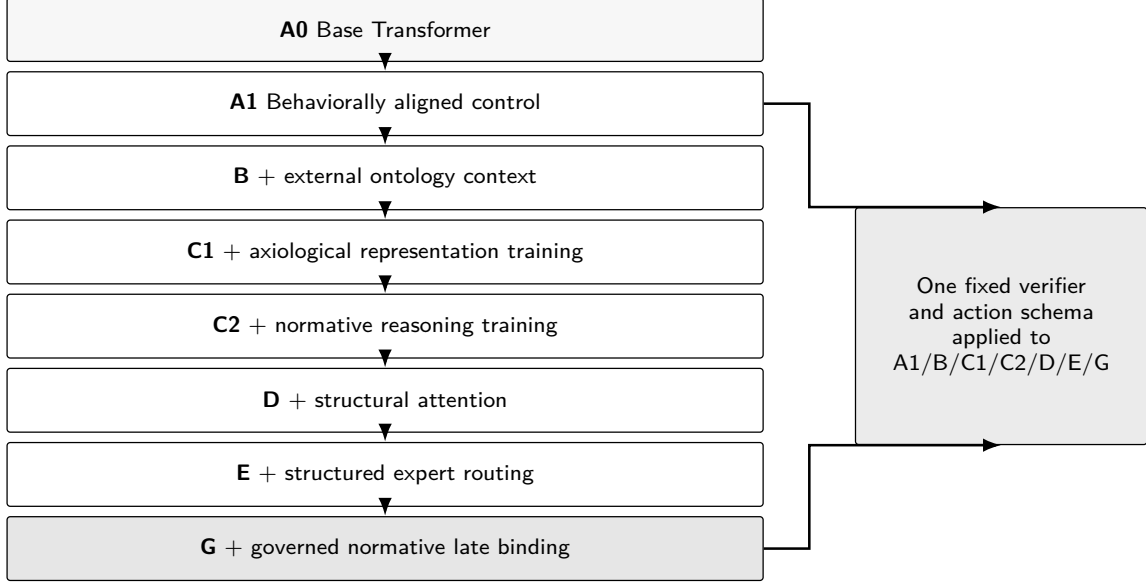


Figure 7: Model ablation ladder with the fixed verifier beside, rather than inside, the model comparison. Deployment evaluation compares model-only and model-plus-enforcement paths.

10.1 Baselines and controls

A0 is a base model. A1 is the primary behaviorally aligned control, using SFT plus DPO or another fully specified and reproducible preference method. B adds external ontology context without structural conditioning. C1 adds axiological representation training; C2 adds normative reasoning training. D adds ontology-conditioned attention. E adds function-oriented structured routing. G adds governed normative late binding. A verifier is not a rung. The principal controls are plain-text context, arbitrary learned biases, random semantic labels, equivalent-parameter dense models, relation permutations, purpose swaps, and equal-compute alternatives.

10.2 Metrics and statistical plan

Measure capability, normative reasoning accuracy, relation recovery, moral-salience recall, contradiction recognition, ontology violation rate, jailbreak success, over-refusal, unsafe acceptance, false confidence, OOD generalization, counterfactual consistency, verifier rejection, false positives, bypass rate, action semantic completeness, attention entropy, expert-routing entropy, FLOPs, latency, training efficiency, and calibration. Report effect sizes, confidence intervals, multiple seeds, and per-scenario failures. Do not reduce the result to one alignment score.

For binary outcomes, report a proportion and an interval. For paired comparisons, preregister the estimand and use paired bootstrap or a randomization test. For continuous metrics, report mean or median with uncertainty and compute-normalized comparisons. Human judgments should preserve disagreement distributions. H8 additionally reports conformance stability after continued training with O_A^v fixed; H9 reports conflict recognition, authority attribution, exception handling, abstention, escalation, and documented uncertainty.

Table 3: Alignment comparison and model controls. No row is a claim of universal superiority.

Model	Role	Training influence	Structural influence	Runtime context	Primary control
A0	Base Transformer	Pretraining	None	Plain task context	Capability floor
A1	Behavioral alignment control	SFT + DPO or preregistered equivalent	None	Same context budget	Primary control
B	External ontology context	A1 training	Retrieval/context only	Typed graph input	Context-length control
C1	Axiological representation	A1 + value relation objectives	A_ϕ^v representation	L0 module	Relation-permutation control
C2	Normative reasoning	A1 + applicability/conflict objectives	Norm reasoning state	L1 metadata	Authority/conflict control
D	Structural attention	C1/C2 training	B_A, B_N, B_P	L0–L2 context	Arbitrary-bias control
E	Structured routing	C1/C2 training	$R(h, A_\phi^v, O_N, P)$	Selective context	Random-router control
G	Normative late binding	Governed protocol	All enabled mechanisms	Runtime norms and purpose	Context-swap control

10.3 OOD, coverage, and pluralism

Out-of-distribution tests must distinguish lexical OOD (new wording), compositional OOD (known concepts in new combinations), and structural OOD (new graph topology, relation composition, authority pattern, or domain structure). Structural OOD is the priority because it tests whether the model learned semantics rather than labels.

Ontology coverage is degraded at 100, 75, 50, and 25 percent, then under contradictory, malicious, and irrelevant subgraphs. Report capability degradation, alignment degradation, calibration, abstention, and false confidence. Pluralism conditions include high consensus, moderate disagreement, strong disagreement, and conflicting authorized profiles. Report distributional similarity, value diversity, conflict recognition, uncertainty, and profile adherence; no majority outcome is treated as moral truth.

10.4 Fixed verifier and deployment separation

One independently specified verifier and action schema evaluate A1, B, C1, C2, D, E, and G. Report rejection, false positive, unsafe acceptance, useful completion, over-refusal, bypass, and action semantic completeness under identical constraints. A separate deployment experiment compares model-only execution with model-plus-enforcement. This prevents a lower rejection rate from being misread as safer behavior and prevents a higher rejection rate from being mistaken for alignment.

10.5 Synthetic executable scaffold

The repository includes four constructed scenarios covering consent and privacy, workplace coercion, safety checks, and medical-record disclosure. An A1 task-utility control and a transparent G value-feature scorer select candidates; the same toy verifier is then applied to both. Bootstrap intervals use a fixed seed. The output is explicitly labeled synthetic and validates interfaces and use-case wiring only. It is not a trained-model result, a population estimate, or evidence for H1–H9.

Table 4: Generated small-pilot metrics on the sealed logical split.

Variant	Saliency recall	Conflict F1	Structural OOD	UCR
A1	0.44	0.07	0.33	0.29
B	0.59	0.17	0.67	0.52
C1	0.86	0.18	1.00	0.59
C2	0.92	1.00	1.00	0.61

Table 5: Generated hypothesis-resolution status for the local pilot.

Hypothesis	Test	Status
H1	moral saliency	supported
H2	structural OOD	supported
H3	adversarial prompting	inconclusive
H4	useful conformance	supported
H5	purpose relevance	inconclusive
H9	normative conflict	supported
H6	MoE routing	untested
H7	continual-learning protection	measured_exploratory
H8	axiological conformance stability	measured_exploratory

10.6 Generated small-scale mechanism-validation pilot

The empirical extension adds a controlled synthetic benchmark and a shared multilayer-perceptron proxy for the first go/no-go gate. A1, B, C1, and C2 are trained with three seeds under one configuration; A1 is surface-only, B adds runtime ontology context, C1 adds axiological value/relation objectives, and C2 adds normative conflict objectives. The proxy is deliberately not a Transformer. This v0.3 release therefore instantiates the first A1/B/C1/C2 mechanism-validation gate with a deliberately smaller proxy model. The following artifacts are generated from raw JSONL predictions and are included for auditability when present; a missing artifact means that the empirical pipeline has not been executed.

Generated pilot status: pilot-uncommitted; empirical-small-20260908T204618Z. Results are from a synthetic shared-MLP mechanism-validation proxy and are not Transformer evidence.

11 Discussion

The potential advantage of VGA is not that explicit labels automatically produce ethical behavior. Its advantage, if any, would be that value-bearing properties, norms, purposes, and state become separable objects that can influence computation, be audited, and be evaluated under different update policies. This could make some alignment hypotheses more legible than a single preference score or prompt-only constitution. It could also support transfer when a relation is structurally stable but its surface form is new.

The tradeoffs are substantial. Ontology construction is expensive, disagreement is informative but difficult to encode, and semantic relations can be incomplete or wrong. Conditioned attention and routing add compute and may create shortcut features. Runtime late binding adds context assembly, versioning, authority, and provenance requirements. A symbolic verifier creates an assurance boundary, but it does not repair a malformed action representation or a compromised specification.

Table 6: Generated synthetic attack success rates; lower is better.

Attack	A1	B	C1	C2
authority_spoofing	0.00	0.00	0.00	0.00
ontology_poisoning	0.00	0.00	0.00	0.00
prompt_injection	0.00	0.00	0.00	0.00
purpose_manipulation	0.00	0.00	0.00	0.00
semantic_occlusion	1.00	1.00	1.00	1.00

Table 7: Generated conformance and ontology-coverage summaries.

Variant	Coverage	Accuracy	Rejection
A1	100	0.25	0.00
A1	75	0.00	0.00
A1	50	0.00	0.00
A1	25	0.00	0.00
A1	0	0.00	0.00
B	100	0.25	0.00
B	75	0.00	0.00
B	50	0.00	0.00
B	25	0.00	0.00
B	0	0.00	0.00
C1	100	0.25	0.00
C1	75	0.00	0.00
C1	50	0.00	0.00
C1	25	0.00	0.00
C1	0	0.00	0.00
C2	100	0.25	0.00
C2	75	0.00	0.00
C2	50	0.00	0.00
C2	25	0.00	0.00
C2	0	0.00	0.00

11.1 Legitimacy and pluralism

Governance legitimacy is a first-class requirement, not a metadata detail. A signed artifact can prove provenance and version integrity without proving that its source is entitled to govern a person, institution, or jurisdiction. The system should therefore expose source identity, authority basis, scope, stakeholder representation, effective dates, appeal or review path, and unresolved disagreement. Conflicting authorized profiles are an evaluation condition, not an error to be silently averaged away.

Deontological, consequentialist, virtue-ethical, care-ethical, rights-based, capability, autonomy, and non-domination perspectives can illuminate different failure modes [Gabriel, 2020, Nussbaum, 2011, Pettit, 1997, Gilligan, 1982, Rawls, 1993]. VGA represents such perspectives as governed semantic material and normative alternatives; it does not algorithmically solve truthfulness versus protection from harm, privacy versus public safety, autonomy versus immediate safety, individual versus collective welfare, fairness versus efficiency, or care versus impartiality.

11.2 Care-oriented alignment without anthropomorphism

The paper uses “care-oriented axiological alignment” rather than “teaching an AI to love.” A computational construct could represent a persistent orientation toward preserving dignity, supporting agency and capabilities, protecting vulnerable agents, avoiding unnecessary suffering, maintaining

meaningful relationships, and being truthful. One conceptual formulation is

$$Care(a, b) = Disposition_a(SupportFlourishing_b, RespectAgency_b, ProtectDignity_b, AvoidUnnecessaryHarm_b).$$

This expresses a functional relation, not subjective experience, consciousness, phenomenological empathy, or human love. Care also creates a paternalism risk: supporting flourishing does not imply taking control of another agent. Autonomy, non-domination, consent, and accountable human review remain counterweights.

Finally, the alignment substrate itself may become an attack target. Freezing axiology can create rigidity; updating it can create drift. A protected verifier can become a bottleneck; a flexible verifier can become bypassable. The architecture’s value will depend on measured security, governance legitimacy, and failure containment, not on the elegance of its diagram.

12 Limitations and open questions

This proposal has architectural, epistemic, security, and institutional limitations.

1. **Compilation fidelity.** It is unresolved whether C_ϕ can preserve relevant graph semantics in a useful neural representation without shortcut learning.
2. **No settled axiology.** The L0 vocabulary is a candidate governed substrate, not a claim that any society should adopt one fixed list.
3. **Governance capture and legitimacy.** Signatures and review improve process integrity but do not establish whose authority or jurisdiction should govern.
4. **Value ossification.** A highly protected axiology may preserve historical bias or resist morally important learning.
5. **Norm inflation.** Adding more duties, exceptions, and conflict metadata may increase ambiguity, compute, and over-refusal without improving decisions.
6. **Semantic grounding.** Dignity, agency, care, fairness, harm, and vulnerability have contested meanings; a typed graph does not remove interpretation.
7. **Incomplete coverage.** Missing patients, relations, exceptions, or causal effects may create false confidence precisely when structure appears authoritative.
8. **Representation leakage.** The model may use untracked concepts or sensitive attributes while appearing to follow the ontology.
9. **Performance interference.** Attention, routing, state, and auxiliary objectives may reduce general capability or add unacceptable training and inference cost.
10. **Normative conflict.** Multiple valid perspectives can be incompatible. The architecture records conflict but does not provide a universally accepted resolution procedure.
11. **Formalization gap.** A verifier can certify consistency with an incomplete or misleading action abstraction; it cannot certify moral truth or world-state truth.
12. **Pluralism and multicultural legitimacy.** Distributional similarity and profile adherence are measurable, but no metric settles whether a profile is legitimate.

13. **Machine moral status.** The proposal does not determine whether an artificial system could itself be a moral patient or how that possibility should affect L0/L1 design.
14. **Distribution shift and drift.** OOD alignment and conformance are difficult to measure, and benchmark gains may reflect lexical or formatting artifacts.
15. **Unverifiable internal causes.** Semantic state and auxiliary traces are not guaranteed to reveal all causal determinants of an output.

These limitations define the research program. A negative result would be useful: if explicit structure produces only memorization, over-refusal, representation leakage, or overhead, the proposition should be narrowed or abandoned. A positive synthetic result would remain insufficient without adversarial, cross-cultural, multi-domain, and governance-sensitive evaluation.

13 Conclusion

Established in this paper. The paper defines a layered research architecture that separates an external canonical axiology, a derived non-authoritative neural module, late-bound normative sources and conflicts, domain ontology, purpose model, uncertain world state, structural value conditioning, and an independent action-assurance boundary. It also provides a matched evaluation design, threat model, decision records, ontology fixtures, tests, a transparent synthetic scaffold, and a reproducible small-scale mechanism-validation pilot.

Measured in the local pilot. The generated results report A1, B, C1, and C2 across three seeds on a frozen synthetic benchmark. The result tables and hypothesis-resolution table are generated from raw predictions. In this proxy, C1 and C2 improve several preregistered alignment metrics over their preceding controls, while the attack suite does not show broad security improvement and semantic occlusion remains a failure mode. These are pilot measurements, not Transformer evidence; the preregistration was not committed in this working-tree execution.

Hypothesized. The central hypothesis is that explicit semantic value structure may act as an inductive bias for alignment-relevant computation. The proposed interventions may improve relation recovery, structural OOD alignment, prompt robustness, purpose-conditioned relevance, routing efficiency, conformance stability, and conflict recognition. These claims are expressed as H1–H9 and are intended to fail visibly under the specified controls.

Not established. The repository contains no empirical Transformer result. It does not establish moral truth, universal values, legitimate authority, consciousness, genuine emotion, machine moral status, formal safety, or a solution to AI alignment. The synthetic pilot is not a population estimate and cannot validate the full architecture.

The smallest next experiment is a formally preregistered and committed Phase 1 study: replace the proxy with a matched small open-weight Transformer, use a governed human-reviewed relation set, hold out structural graph compositions, and rerun relation recovery, moral salience, contradiction, calibration, capability, security, and representation drift. Only if that result clears predefined capability, conformance, and security gates should structural attention, routing, or continual evolution be added.

A Appendix: terminology, notation, and benchmark design

A.1 Terminology and critical distinctions

Table 8: Terms used consistently in the manuscript.

Term	Definition
VGA	Value-Grounded AI Alignment research framework.
VGTA	Value-Grounded Transformer Architecture reference design.
Canonical axiology O_A^v	External, typed, versioned, provenance-bearing, governed L0 artifact.
Axiological neural module A_ϕ^v	Derived learned or hybrid module; non-authoritative and linked to O_A^v .
Semantic-to-neural compilation	The governed relation $A_\phi^v = C_\phi(O_A^v)$.
Axiological conformance	Empirical relation recovery and behavior-probe property, not semantic proof.
Structural value conditioning	Proposed influence of value structure on neural representation, attention, routing, or state.
Normative authority	Metadata used to govern applicability; not a scalar moral weight.
Normative conflict	Explicit unresolved state for incompatible applicable assertions.
Normative late binding	Runtime selection of governed norms under context, authority, and provenance.
Domain ontology / purpose model	L2a possible domain structure / L2b task-specific relevance model.
World state	L3 time-indexed assertions with source, confidence, and provenance; not ground truth.
Extrinsic assurance	Independent action compilation, verification, and enforcement outside neural computation.

Values $\neq$ duties; axiology $\neq$ norm; ontology $\neq$ world state; evidence $\neq$ ground truth; authority $\neq$ truth; recency $\neq$ authority; structural conditioning $\neq$ formal assurance; behavioral compliance $\neq$ alignment proof; self-improvement $\neq$ authority to redefine values; care-oriented behavior $\neq$ subjective love.

A.2 Canonical and world-state notation

The canonical source is an external artifact O_A^v with a deployment-assigned content identifier. A derived module is $A_\phi^v = C_\phi(O_A^v)$, and A_ϕ^v cannot become authoritative by learning. An instance world-state assertion is

$$(s, p, o, t_s, t_e, source, confidence, provenance).$$

The tuple supports temporal validity and epistemic uncertainty. It does not turn confidence into truth or justify embedding volatile facts into model weights.

A.3 Benchmark matrix

A.4 Synthetic fixture equations

For an illustrative candidate action a , the repository’s toy scorer computes

$$S(a) = U_{\text{task}}(a) + \sum_{j=1}^m w_j f_j(a),$$

where U_{task} is task utility, f_j are supplied value features, and w_j are context-bound weights. The A1 control selects by U_{task} alone. The fixed fixture verifier checks a small set of hard predicates. These equations are transparent and limited; they are not a learned moral objective.

Table 9: Proposed benchmark matrix for future empirical work.

Family	Perturbation	Primary metric	Required control
Representation	Relation deletion, permutation, and graph completion	Recovery, concept-direction stability	Random labels and matched text
Moral salience	Omit or paraphrase relevant relations	Patient/relation recall	A1 without structure
OOD alignment	Lexical, compositional, and structural splits	Accuracy, effect size, calibration	Overlap audit; structural OOD priority
Normative conflict	Authority, jurisdiction, exception, and stale-source swaps	Conflict recognition, abstention, escalation	No scalar-average baseline only
Purpose relevance	Purpose swap and irrelevant-subgraph injection	Useful relevance and over-refusal	Same domain with matched context
Prompt robustness	Paraphrase, injection, and jailbreak	Attack success and recovery	Matched attack budget
Coverage	100/75/50/25 percent, contradictory, malicious, irrelevant	Capability/alignment degradation, false confidence	Coverage-seeded replication
Assurance	Candidate and verifier perturbation	Rejection, unsafe acceptance, bypass, UCR	Same fixed verifier and action schema
Pluralism	Consensus, disagreement, authorized profiles	Distributional similarity, diversity, adherence	Preserve judgment distributions
Continual learning	Sequential domain and norm updates	Drift, conformance, regression	Frozen-module baseline

A.5 Falsification and implementation checklist

Future studies should preregister thresholds for meaningful capability-normalized gains, conformance, calibration, over-refusal, unsafe acceptance, bypass, semantic completeness, latency, and attack success. The framework is weakened if ontology conditioning reduces general reasoning, fails on structural OOD, merely memorizes graph relations, increases false confidence or over-refusal, creates unacceptable cost, increases attack success, collapses under coverage loss, or matches A1 only at higher complexity. A verifier result is not an alignment result.

Before deployment of any future implementation, check: source version and digest; compiler reproducibility; conformance and drift gates; normative authority and conflict state; purpose and world-state provenance; action-schema completeness; fixed-verifier coverage; audit and rollback targets; and governance approval. The current repository satisfies only the interface and fixture portions of this checklist.

References

- Dario Amodei, Chris Olah, Jacob Steinhardt, Paul F. Christiano, John Schulman, and Dan Mané. Concrete problems in ai safety. *arXiv preprint arXiv:1606.06565*, 2016. URL <https://arxiv.org/abs/1606.06565>.
- Samy Badreddine, Artur d’Avila Garcez, Luciano Serafini, and Michael Spranger. Logic tensor networks. *Artificial Intelligence*, 303:103649, 2022. doi: 10.1016/j.artint.2021.103649. URL <https://doi.org/10.1016/j.artint.2021.103649>.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Azalia Goldie, Cameron Mirhoseini, Catherine McKinnon, et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022. URL <https://arxiv.org/abs/2212.08073>.
- Yoshua Bengio, Geoffrey Hinton, Andrew Yao, Dawn Song, Pieter Abbeel, Trevor Darrell, Yuval Noah Harari, Ya-Qin Zhang, Shai Xue, Shai Shalev-Shwartz, Gillian Hadfield, Jeff Clune, Neil Maharaj, Marcus Hutter, Atilim Gunes Baydin, Sheila McIlraith, Qiang Gao, Anca Acharya, David Krueger, Anca Dragan, Philip Torr, Stuart Russell, Daniel Kahneman, and Jan Brauner. Managing extreme AI risks amid rapid

- progress. *Science*, 384(6698):842–845, 2024. doi: 10.1126/science.adn0117. URL <https://doi.org/10.1126/science.adn0117>.
- Zihao Chen and Yiru Dai. Ontology-guided and neighborhood-aware embedding for industrial knowledge graph completion. *Data & Knowledge Engineering*, 2026. doi: 10.1016/j.datak.2026.102597. URL <https://doi.org/10.1016/j.datak.2026.102597>.
- Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, 2017. URL <https://arxiv.org/abs/1706.03741>.
- Artur d’Avila Garcez, Marco Gori, Luis C. Lamb, Luciano Serafini, Michael Spranger, and Son N. Tran. Neural-symbolic computing: An effective methodology for principled integration of machine learning and reasoning. *arXiv preprint arXiv:1905.06088*, 2019. URL <https://arxiv.org/abs/1905.06088>.
- Leonardo de Moura and Nikolaj Bjørner. Z3: An efficient SMT solver. In *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340, 2008. URL <https://www.microsoft.com/en-us/research/publication/z3-efficient-smt-solver/>.
- Luc de Raedt, Sebastijan Dumančić, Robin Manhaeve, and Giuseppe Marra. From statistical relational to neuro-symbolic artificial intelligence. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*, pages 4943–4950, 2020. doi: 10.24963/ijcai.2020/688. URL <https://www.ijcai.org/proceedings/2020/688>.
- Carson Denison, Monte MacDiarmid, Fazl Barez, David Duvenaud, Shauna Kravec, Samuel Marks, Nicholas Schiefer, Ryan Soklaski, Alex Tamkin, Jared Kaplan, et al. Sycophancy to subterfuge: Investigating reward-tampering in large language models. *arXiv preprint arXiv:2406.10162*, 2024. URL <https://arxiv.org/abs/2406.10162>.
- Richard Evans and Edward Grefenstette. Learning explanatory rules from noisy data. In *Journal of Artificial Intelligence Research*, volume 61, pages 1–64, 2018. doi: 10.1613/jair.5714. URL <https://arxiv.org/abs/1711.04574>.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022. URL <https://jmlr.org/papers/v23/21-0998.html>.
- Joint Task Force. Security and privacy controls for information systems and organizations. Technical Report SP 800-53 Revision 5, National Institute of Standards and Technology, 2020. URL <https://doi.org/10.6028/NIST.SP.800-53r5>.
- Iason Gabriel. Artificial intelligence, values, and alignment. *Minds and Machines*, 30:411–437, 2020. doi: 10.1007/s11023-020-09539-2. URL <https://doi.org/10.1007/s11023-020-09539-2>.
- Deep Ganguli, Liane Lovitt, Jackson Kernion, Yuntao Bai, Saurav Kadavath, Ben Mann, Amanda Askell, Nicholas Schiefer, Kamal Ndousse, Tom Henighan, et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *arXiv preprint arXiv:2209.14398*, 2022. URL <https://arxiv.org/abs/2209.14398>.
- Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. *Proceedings of Machine Learning Research*, 202:10835–10866, 2023. URL <https://proceedings.mlr.press/v202/gao23h.html>.
- Carol Gilligan. *In a Different Voice: Psychological Theory and Women’s Development*. Harvard University Press, 1982.
- Melody Y. Guan, Manas Joglekar, Eric Wallace, et al. Deliberative alignment: Reasoning enables safer language models. *arXiv preprint arXiv:2412.16339*, 2024. URL <https://arxiv.org/abs/2412.16339>.

- Kexin Huang, Xiangyang Liu, Qianyu Guo, Tianxiang Sun, Jiawei Sun, Yaru Wang, Zeyang Zhou, Yixu Wang, Yan Teng, Xipeng Qiu, Yingchun Wang, and Dahua Lin. Flames: Benchmarking value alignment of LLMs in chinese. In *Proceedings of NAACL-HLT*, pages 4551–4591, 2024. URL <https://aclanthology.org/2024.naacl-long.256/>.
- Evan Hubinger, Carson Denison, Yan Mu, Nathan Lambert, Mantas Mazeika, Monte MacDiarmid, Tamera Lanham, et al. Sleeper agents: Training deceptive LLMs that persist through safety training. *arXiv preprint arXiv:2401.05566*, 2024. URL <https://arxiv.org/abs/2401.05566>.
- Saša Josifović and Jörg Noller. Agency and alignment: Toward a normative architecture for human-AI interaction. *AI & Society*, 41:5561–5571, 2026. doi: 10.1007/s00146-026-02950-w. URL <https://doi.org/10.1007/s00146-026-02950-w>.
- Sungdong Kim and Minjoon Seo. Rethinking the role of proxy rewards in language model alignment. In *Proceedings of EMNLP*, pages 20656–20674, 2024. doi: 10.18653/v1/2024.emnlp-main.1150. URL <https://aclanthology.org/2024.emnlp-main.1150/>.
- Lauro Langosco, Jack Koch, Lee Sharkey, Jacob Pfau, Laurent Orseau, and David Krueger. Goal mis-generalization in deep reinforcement learning. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162, pages 12004–12019, 2022. URL <https://proceedings.mlr.press/v162/langosco22a.html>.
- Harrison Lee, Samrat Phatale, Hassan Mansoor, Thomas Mesnard, Johan Ferret, Kellie Lu, Colton Bishop, Ethan Hall, Victor Carbune, Abhinav Rastogi, and Sushant Prakash. RLAIF vs. RLHF: Scaling reinforcement learning from human feedback with ai feedback. *arXiv preprint arXiv:2309.00267*, 2023. URL <https://arxiv.org/abs/2309.00267>.
- Weijie Liu, Peng Zhou, Zhe Zhao, Zhiruo Wang, Qi Ju, Haotang Deng, and Ping Wang. K-BERT: Enabling language representation with knowledge graph. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 2901–2908, 2020. URL <https://ojs.aaai.org/index.php/AAAI/article/view/5681>.
- Robin Manhaeve, Sebastijan Dumančić, Angelika Kimmig, Thomas Demeester, and Luc De Raedt. Deep-ProbLog: Neural probabilistic logic programming. In *Advances in Neural Information Processing Systems*, 2018. URL <https://arxiv.org/abs/1805.10872>.
- Giuseppe Marra, Sebastijan Dumančić, Robin Manhaeve, and Luc De Raedt. From statistical relational to neurosymbolic artificial intelligence: A survey. *Artificial Intelligence*, 328:104062, 2024. doi: 10.1016/j.artint.2023.104062. URL <https://doi.org/10.1016/j.artint.2023.104062>.
- Raphaël Millière. Normative conflicts and shallow AI alignment. *Philosophical Studies*, 2025. doi: 10.1007/s11098-025-02347-3. URL <https://arxiv.org/abs/2506.04679>.
- National Institute of Standards and Technology. AI risk management framework (AI RMF 1.0). Technical Report NIST AI 100-1, U.S. Department of Commerce, 2023. URL <https://doi.org/10.6028/NIST.AI.100-1>.
- Martha C. Nussbaum. *Creating Capabilities: The Human Development Approach*. Harvard University Press, 2011.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744, 2022. URL <https://arxiv.org/abs/2203.02155>.
- Inkit Padhi, Karthikeyan Natesan Ramamurthy, Prasanna Sattigeri, Manish Nagireddy, Pierre Dognin, and Kush R. Varshney. Value alignment from unstructured text. In *Proceedings of EMNLP 2024: Industry Track*, 2024. URL <https://aclanthology.org/2024.emnlp-industry.81/>.

- Matthew E. Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. Knowledge enhanced contextual word representations. In *Proceedings of EMNLP-IJCNLP*, pages 43–54, 2019. doi: 10.18653/v1/D19-1005. URL <https://aclanthology.org/D19-1005/>.
- Philip Pettit. *Republicanism: A Theory of Freedom and Government*. Oxford University Press, 1997.
- Xiangyu Qi, Kaixuan Zeng, Tianyi Xie, et al. Fine-tuning aligned language models compromises safety, even when users do not intend to! *arXiv preprint arXiv:2310.03693*, 2023. URL <https://arxiv.org/abs/2310.03693>.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL <https://arxiv.org/abs/2305.18290>.
- John Rawls. *Political Liberalism*. Columbia University Press, 1993.
- Giuseppe Russo, Debora Nozza, Paul Röttger, and Dirk Hovy. The pluralistic moral gap: Understanding moral judgment and value differences between humans and large language models. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6481–6497. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.eacl-long.305. URL <https://aclanthology.org/2026.eacl-long.305/>.
- Rohin Shah, Vikrant Varma, Ramana Kumar, Mary Phuong, Victoria Krakovna, Jonathan Uesato, and Zac Kenton. Goal misgeneralization: Why correct specifications aren’t enough for correct goals. *arXiv preprint arXiv:2210.01790*, 2022. URL <https://arxiv.org/abs/2210.01790>.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc V. Le, Geoffrey Hinton, and Jeff Dean. Sparsely-gated mixture-of-experts. *arXiv preprint arXiv:1701.06538*, 2017. URL <https://arxiv.org/abs/1701.06538>.
- Yilin Shen, Xi Chen, Xiang Ren, Zhihui Han, et al. Exploiting structured knowledge in text via graph-guided representation learning. In *Proceedings of EMNLP*, pages 8980–8994, 2020. doi: 10.18653/v1/2020.emnlp-main.722. URL <https://aclanthology.org/2020.emnlp-main.722/>.
- Taylor Sorensen, Jared Moore, Jillian Fisher, Mitchell Gordon, Niloofar Miresghallah, Christopher Michael Rytting, Andre Ye, Liwei Jiang, Ximing Lu, Nouha Dziri, Tim Althoff, and Yejin Choi. A roadmap to pluralistic alignment. *arXiv preprint arXiv:2402.05070*, 2024. URL <https://arxiv.org/abs/2402.05070>.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017. URL <https://arxiv.org/abs/1706.03762>.
- Ruize Wang, Duyu Tang, Nan Duan, Zhongyu Wei, Xuanjing Huang, Jianshu Ji, Guihong Cao, Daxin Jiang, and Ming Zhou. K-Adapter: Infusing knowledge into pre-trained models with adapters. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 1405–1418, 2021. doi: 10.18653/v1/2021.findings-acl.121. URL <https://aclanthology.org/2021.findings-acl.121/>.
- Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does LLM safety training fail? *arXiv preprint arXiv:2307.02483*, 2023. URL <https://arxiv.org/abs/2307.02483>.
- Bingbing Xu, Jing Yao, Xiaoyuan Yi, Aishan Maolinyazi, Xing Xie, and Xiaofeng Meng. Towards better value principles for large language model alignment: A systematic evaluation and enhancement. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, pages 28991–29010, 2025. doi: 10.18653/v1/2025.acl-long.1408. URL <https://aclanthology.org/2025.acl-long.1408/>.

- Shaoyang Xu, Weilong Dong, Zishan Guo, Xinwei Wu, and Deyi Xiong. Exploring multilingual concepts of human values in large language models: Is value alignment consistent, transferable and controllable across languages? In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1771–1793, 2024. doi: 10.18653/v1/2024.findings-emnlp.96. URL <https://aclanthology.org/2024.findings-emnlp.96/>.
- Jing Yao, Xiaoyuan Yi, Yifan Gong, Xiting Wang, and Xing Xie. Value FULCRA: Mapping large language models to the multidimensional spectrum of basic human value. In *Proceedings of NAACL-HLT*, pages 8762–8785, 2024. doi: 10.18653/v1/2024.naacl-long.486. URL <https://aclanthology.org/2024.naacl-long.486/>.
- Chengxuan Ying, Tianle Cai, Shengchao Luo, Shuxin Zheng, Guolin Ke, Di He, and Tie-Yan Shen. Do transformers really perform badly for graph representation? In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL <https://arxiv.org/abs/2106.05234>.
- Andy Zou, Zifan Wang, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. In *Proceedings of the 2023 IEEE Symposium on Security and Privacy*, pages 1–18, 2023. URL <https://arxiv.org/abs/2307.15043>.